Supporting Information

Definitional Instability in Kinase Inhibitor Selectivity: An Empirical Characterization and Motivated Properties for a Well-Formed Measure

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This document reports, in full, four robustness checks summarized in the main text, the variants of the candidate measure (Section S5), and further analyses summarized briefly in the main text (Section S6). All analyses use the curated panels of the main text, one reference target per human protein kinase, except where Section S6 compares them with the released panels. All numbers are reproducible from the code repository (`https://github.com/polinavino/kinase-selectivity-definitions`); the script generating each table is named in its caption, and Table S7 lists the script behind every result reported in the main text.

Table S1: Panel-size recoverability under alternative correlation bars

The main text writes p^* for the smallest panel size at which mean Spearman r against the full panel exceeds 0.90, a bar borrowed from the "excellent" range of intraclass-correlation guidelines (Results). Table S1 reports p^* at four bars, including the main-text bar of 0.90, for all four datasets.

The ratio is the slowest to recover, or tied for slowest, on every dataset for every bar tested, and the Gini coefficient the second-slowest on Davis, Klaeger, and Anastassiadis. On Metz it ties the entropy at every bar. No bar in $[0.80, 0.95]$ reverses the order of two definitions on any dataset. The bar decides only whether some of them tie.

Table S1: Panel-size p^* (targets) at four correlation bars. (`pstar_bar_sensitivity.py`)

Davis (363 targets)	entropy	gini	s_score	ratio
bar = 0.80	50	80	50	320
bar = 0.85	50	110	50	320
bar = 0.90	50	110	50	350
bar = 0.95	50	170	80	350

Klaeger (219 targets)	entropy	gini	s_score	ratio
bar = 0.80	50	170	50	200
bar = 0.85	50	200	50	200
bar = 0.90	80	200	80	219
bar = 0.95	110	219	140	219

Anastassiadis (291 targets)	entropy	gini	s_score	ratio
bar = 0.80	50	170	50	260
bar = 0.85	50	170	50	260
bar = 0.90	50	200	50	260
bar = 0.95	80	260	80	290

Metz (172 targets)	entropy	gini	s_score	ratio
bar = 0.80	50	50	50	170
bar = 0.85	50	50	50	170
bar = 0.90	80	80	50	170
bar = 0.95	110	110	80	170

Table S2: Bosc et al. window/ranking score and the CATDS proxy

Bosc et al.'s window score (WS) and ranking score (RS) and a CATDS-style proxy (Discussion) were computed on all four datasets. RS at rank $k + 1$ is $R(k)$ apart from the floor fallback (Methods), so its agreement with the ratio follows from its definition. (`bosc_catds_comparison.py`)

Table S2: Spearman correlation of WS/RS against the ratio family, and of the CATDS proxy against the ratio, entropy, and Gini families.

Dataset	n_{cpd}	WS vs. ratio	RS vs. ratio	CATDS vs. ratio	CATDS vs. entropy / Gini
Davis	68	0.932	0.904	0.636	0.977 / 0.977
Klaeger	222	0.920	0.888	0.725	0.635 / 0.982
Anastassiadis	178	0.957	0.883	0.698	0.945 / 0.964
Metz	704	0.910	0.899	0.447	0.883 / 0.906

WS and RS correlate strongly with the ratio family on every dataset ($r = 0.88$–0.96). The CATDS proxy correlates most strongly with the Gini on every dataset, and more with the entropy than with the ratio on every dataset except Klaeger. The proxy does not reproduce the concentration dependence of the published CATDS score.

Table S3: Cross-definition correlations under the original metric operationalizations

Methods reports summary statistics for this check; the full 4×4 correlation matrices are given here. (`original_operationalization_robustness.py`)

Table S3: Davis ($n = 68$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (complete matrix).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.344	0.963	0.563
entropy(literal)	0.344	1.000	0.368	0.688
Gini	0.963	0.368	1.000	0.510
ratio	0.563	0.688	0.510	1.000

Klaeger ($n = 222$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured high-confidence entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.567	0.511	0.085
entropy(literal)	0.567	1.000	0.532	0.661
Gini	0.511	0.532	1.000	0.657
ratio	0.085	0.661	0.657	1.000

Metz ($n = 704$): manuscript S-score/Gini/ratio vs. literal Ka-weighted entropy (measured entries only, per compound).

	S-score	entropy(literal)	Gini	ratio
S-score	1.000	0.291	0.821	0.193
entropy(literal)	0.291	1.000	0.385	0.758
Gini	0.821	0.385	1.000	0.155
ratio	0.193	0.758	0.155	1.000

Anastassiadis ($n = 178$): manuscript S-score/entropy/ratio vs. native %inhibition Gini (no baseline subtraction).

	S-score	entropy	Gini(native)	ratio
S-score	1.000	0.880	0.340	0.478
entropy	0.880	1.000	0.629	0.524
Gini(native)	0.340	0.629	1.000	0.326
ratio	0.478	0.524	0.326	1.000

On Davis, Klaeger and Metz, the literal entropy correlates more strongly with the ratio than with either the S-score or the Gini. Under this operationalization, the entropy would not be classified in the distribution-based family on any of the three affinity datasets. The native-Gini check on Anastassiadis preserves the pattern only partly: the native Gini correlates with the entropy (0.63)

but weakly and about equally with the S-score (0.34) and the ratio (0.33).

Table S4: Candidate measure cross-dataset validation

Table S4 reports three checks for each dataset. Orientation is the correlation with the number of active kinases, where more negative means more correctly oriented toward "fewer active targets = more selective". D2 is the percentage of pairs of scored compounds whose order reverses somewhere in $B = [x_0, \tau^*]$ (Methods). It is given for the emphasis parameter β with the gate x_0 fixed (the featured measure), for the gate x_0 itself swept as a free parameter, and for the entropy's own baseline. D3 is the percentage of compounds for which the score does not rise when one kinase is added at the activity threshold τ^*, the strongest binding D3 allows and so the addition most likely to raise the score. B is $[5.0, 6.0]$ on Davis and Klaeger, $[4.0, 6.0]$ on Metz, and $[20, 50]$% inhibition on Anastassiadis, and the smoothing width is $T = \tau^* - x_0$ on every dataset (Methods). (`candidate_validation.py`, `d3_pair_reversals.py`)

Table S4: Candidate measure: orientation, D2 (pairs reversing in B, %), and D3, across all four datasets, over the n compounds that D1 allows to be scored.

Dataset	n	orientation r	D2, gate fixed	D2, gate swept	D2, entropy	D3 (%)
Davis	68	−0.968	0.44	9.22	7.90	100.0
Klaeger	200	−0.948	0.42	12.73	12.30	100.0
Anastassiadis	133	−0.934	0.36	14.24	10.58	100.0
Metz	699	−0.783	0.72	23.59	22.80	100.0

Swept over the same interval, the candidate's gate reverses about as many pairs as the entropy's baseline does, and the emphasis parameter, with the gate fixed, reverses fewer than one pair in a hundred. Removing the gate inverts the measure's orientation entirely (gated $r = -0.948$ vs. ungated $r = +0.91$ against n_{active} on Klaeger), which is why the gate is required for correct orientation.

S5: Variants of the candidate measure

The featured candidate is one member of a family of measures. Three axes of variation are reported here: the floor, the Rényi order, and the smoothing width T. D3 is proved only for the featured measure (Shannon order, $\beta = x_0$, $T \geq 0.58\,(\tau^* - x_0)$), and is otherwise checked by adding one kinase at τ^* to every compound. (`family_construction.py`, `family_panelsize.py`, `candidate_validation.py`)

Quantile floor. Replacing the fixed floor x_0 with a per-compound low quantile of the profile (the 5th to 15th percentile) gives a variant that is invariant to shifts of the readout, but on these data it is not a separate test: on Davis and Klaeger every such percentile equals the floor for all but at most two compounds, so the variant ranks compounds exactly as the featured measure does (Spearman 1.000).

Rényi order. Generalizing the Shannon functional to Rényi order $\alpha > 0$,

$$-H_\alpha(p) = \frac{1}{\alpha - 1} \log_2 \sum_i p_i^\alpha,$$

gives a family that recovers the featured measure at $\alpha = 1$. D1 holds for every order, since the gate is applied before the entropy is computed. D3 holds on every compound with one kinase added at τ^* for each order tested. The emphasis baseline's D2 separates the orders (Table S5): the orders 0.5, 1 and 2 pass on both datasets, and $\alpha = 3$ fails on both.

Table S5: Rényi-order variants on Davis and Klaeger: D2 for the emphasis baseline (pairs reversing in B, %), agreement with the Shannon ranking (Spearman), and panel size p^* on Klaeger (smallest panel reaching mean $r > 0.90$). On Metz every order also reaches $p^* = 80$. (`family_construction.py`, `family_panelsize.py`)

Order	D2, Davis	D2, Klaeger	Spearman vs. $\alpha = 1$	p^*, Klaeger
$\alpha = 0.5$	0.40	0.12	0.999–1.000	80
$\alpha = 1$	0.44	0.42	—	80
$\alpha = 2$	0.92	0.96	0.999–1.000	80
$\alpha = 3$	1.14	1.59	0.998–0.999	80

The order does not affect panel recoverability. Every order tested, from 0.01 to 3, reaches $p^* = 80$ on both Klaeger and Metz, and the mean convergence curves differ by at most 0.003 between orders at every panel size. We feature $\alpha = 1$ because it is the Shannon entropy, for which D3 is proved.

Smoothing width. The featured measure uses $T = \tau^* - x_0$. On Klaeger, over $T = 0.5$, 0.75, 1, 1.5, and 2 (in units of $\tau^* - x_0 = 1$), orientation is unchanged ($r \approx -0.95$) and the rankings agree closely across all five values (worst-case pairwise Spearman 0.999). The emphasis baseline's D2 passes for $T \geq 0.75$ (pairs reversing: 0.76%, 0.42%, 0.11%, 0.03% for $T = 0.75$, 1, 1.5, 2) and fails at $T = 0.5$ (1.81%), which also falls below the D3 bound $T \geq 0.58$. As $T \to 0$ the softplus converges to the hard cutoff, recovering the entropy, with its baseline sensitivity and D3 failure.

S6: Additional analyses

The analyses below are summarized briefly in the main text, and are given here in the order the main text cites them.

Released panels. The main analyses use one reference target per human protein kinase (Methods). Table S6 compares the main Davis and Klaeger results with the released panels: Davis's 433 constructs and Klaeger's 343 Kinobeads targets. On Davis, the best affinity per gene is also given as an alternative to the reference construct. On the released Davis panel, 26 of the 68 compounds have their two best-bound constructs from the same gene. The most ratio-unstable compound there, GDC-0879, owes its 0.023 top_1–top_2 gap to BRAF(V600E) and BRAF, and its gap to the next gene (RAF1) is 1.54. The released Klaeger panel contains 120 proteins that are not protein kinases: 46 subunits or adaptors of kinase complexes (for example the cyclins CCNA2–CCNT2, CKS1B, the AMPK and PKA regulatory subunits, and the AP2 adaptor subunits), 10 small-molecule and lipid kinases, 20 other enzymes (for example COMT, NQO2, TOP2B, PYGB, PYGL, and PTPN1), and 44 other proteins, including ribosomal proteins and SLC transporters. Kinobeads co-capture complex partners with their kinases, so a CDK hit can also count its cyclin. These columns hold 519 of the 2,763 active entries (18.8%), in 117 of the 222 compounds. Six compounds are active only on them (AZD-2014, AZD-8186, NMS-1286937, SGX-523, TG-100115, and volitinib) and have no active target on the curated panel. Four further columns are protein groups (CSNK2A1;CSNK2A3, PDPK1;PDPK2P, PRKCE;PRKCH, and PRKX;PRKY) and are left out of the curated panel. Most results change little between the panels. The largest changes are on Davis, where the ratio agrees more with the other definitions on the curated panel and its dependence on the number of active targets and on the second-ranked affinity disappears, and on Klaeger, where with all compounds included the Gini sits closer to the ratio on the curated panel.

Further candidate features. Five further features were tested against within-family instability on Klaeger and Davis: two concentration measures from industrial-organization economics, the field the Gini coefficient comes from (the top-target concentration ratio CR_1 and the Herfindahl–Hirschman Index, HHI), the primary-target affinity, a near-tie count generalizing the top_1–top_2 gap, and the active-affinity coefficient of variation. CR_1 and HHI are near-collinear with n_{active} ($r = -0.98$ and -0.997 to -0.998 on both datasets) and so carry the same information. The other three were weaker than the strongest feature of main-text Table 2 in every family and on both datasets. (`instability_predictor_candidates.py`)

Klaeger confidence filter. The high-confidence filter removes 793 of the 5,916 drug–target rows in the table (13.4%), one of which has no numeric K_d^{app}. A removed pair is treated identically to one with no detected binding and assigned the same $\text{p}K_d = 5.0$ background value. Of the 792 low-confidence pairs with a numeric K_d^{app}, 406 involve curated targets, and 403 of these fall within the 222×219 matrix. Adding the 403 pairs changes 0.83% of its entries, shifts every cell of the Klaeger triangle of main-text Table 1 by at most 0.08, and leaves every Klaeger panel-size p^* value unchanged. Retaining all low-confidence pairs also adds 7 compounds and shifts the Klaeger cells of Table 1 by at most 0.06. (`klaeger_filter_sensitivity.py`)

Skewness and Gini instability. Among the 150 Klaeger compounds with at least three active kinases, skewness correlates with the share of a compound's weight carried by kinases that the β sweep crosses ($r = 0.37$), and that share is associated with Gini instability with the number of

Table S6: Main Davis and Klaeger results on the released and curated panels. Correlations are Spearman r; "active" rows re-rank the compounds with an active kinase among themselves. (`davis_gene_level.py`, `klaeger_protein_kinases.py`)

Davis	Released (433)	Curated (363)	Best per gene (363)
Same-gene top two constructs	26	0	0
Table 1: S–E, S–G, E–G	0.946, 0.952, 0.999	0.963, 0.963, 0.999	0.960, 0.960, 1.000
Table 1: R against S, E, G	0.456, 0.343, 0.343	0.563, 0.513, 0.510	0.511, 0.478, 0.474
Table 2: n_{active}, entropy	-0.279	-0.301	-0.268
Table 2: n_{active}, ratio	$+0.294$	$+0.007$	$+0.189$
Table 2: top_1–top_2 gap, ratio	-0.444	-0.445	-0.566
Table 2: top_2 pK_d, ratio	$+0.296$	$+0.147$	$+0.203$
Candidate vs. entropy	0.999	0.997	0.998
p^*: candidate, entropy, S, G, R	50, 50, 50, 110, 410	50, 50, 50, 110, 350	50, 50, 50, 110, 350

Klaeger	Released (343)	Curated (219)
Zero-active compounds	16	22
Table 1, all: S–E, S–G, E–G	0.910, 0.722, 0.840	0.898, 0.511, 0.657
Table 1, all: R against S, E, G	0.223, 0.455, 0.616	0.085, 0.344, 0.657
Table 1, active: S–E, S–G, E–G	0.913, 0.911, 0.998	0.901, 0.897, 0.997
Table 1, active: R against S, E, G	0.418, 0.621, 0.634	0.391, 0.621, 0.637
$r(\text{S},\text{G}) - r(\text{G},\text{R})$, active, 95% interval	$[0.19, 0.37]$	$[0.17, 0.36]$
Pooled rank σ, zero-active vs. active	74.0 vs. 31.2	78.0 vs. 32.1
Table 2: n_{active}, entropy	-0.381	-0.369
Table 2: top_1–top_2 gap, ratio	-0.341	-0.333
Table 2: active range, S-score	-0.354	-0.370
Table 2: skewness, Gini	$+0.346$	$+0.277$
Candidate vs. entropy	0.998	0.998
p^*: candidate, entropy, S, G, R	80, 80, 110, 290, 343	80, 80, 80, 200, 219

active kinases held fixed (partial $r = 0.71$). Holding it fixed as well lowers the skewness association from 0.51 to 0.29, so this mechanism accounts for part of the association. (`claim_checks.py`, section 16)

Type 3 compounds. Seven of the ten largest entropy–ratio disagreements among active Klaeger compounds have a top_1–top_2 gap above 1, for example dovitinib ($n_{\text{active}} = 22$, gap = 1.93) and BMS-690514 ($n_{\text{active}} = 20$, gap = 1.98). The ratio ranks these seven 14th to 56th of 222, and the entropy and Gini rank them 120th to 195th. (`claim_checks.py`, section 5; `klaeger_selectivity_results.csv`)

Most stable compounds. On Klaeger the most stable compounds are among the most promiscuous. XL-228 ($\sigma = 8.5$) and PF-03814735 ($\sigma = 9.9$), with 40–50 active kinases, have median ranks of 199–218.5 of 222 under every definition, and mean instability falls from about 30–35 for compounds with 1 to 40 active kinases to 19 above 40. On Davis, compounds with few active kinases are the most stable on average (mean $\sigma = 3.9$ for one or two, 4.5 for three to five, and 7.3 for more than 40). The most stable, nintedanib (main-text Figure 4, left, $\sigma = 2.1$), has 153 active targets and median ranks of 59–62 of 68 under the four definitions. Without the ratio's configurations, the mean instability of the 32 Davis compounds with more than 40 active targets falls from 7.3 to 3.7.

(`claim_checks.py`, sections 7 and 18)

Table S7: Provenance of reported results

Every result in the main text is printed by a script in the code repository. Table S7 gives, for each result, the script that computes it. Each script saves its printed output as `outputs/<script>.txt`; numbers in parentheses are numbered sections of that output, and a few results are read from the CSV file the script writes, named in parentheses. Random steps are seeded, so rerunning a script reproduces its output.

Table S7: Provenance of reported results, grouped by manuscript section.

Location	Result	Script (output section)
Methods, Datasets	Curated panels: reference targets, removed constructs and proteins	`curate_panels.py`
Methods, Datasets; Section S6	Released Davis constructs against the curated panel (Table S6)	`davis_gene_level.py`
Methods, Datasets; Section S6	Released Klaeger panel against the curated panel (Table S6)	`klaeger_protein_kinases.py`
Methods, Datasets; Candidate	Active fractions; stricter S-score cutoff on Davis; missing entries and noise floor on Anastassiadis; Metz coverage per compound	`dataset_descriptive_stats.py`
Methods, Datasets	Active fractions of the 46 compounds shared by Davis and Klaeger	`claim_checks.py` (9)
Methods, Datasets; Section S6	Klaeger confidence filter: rows removed, relaxed-filter sensitivity	`klaeger_filter_sensitivity.py`
Methods, Datasets	Metz coverage distribution	`metz_coverage.py`
Methods, Datasets	Metz rankings vs number of measured kinases	`claim_checks.py` (10)
Methods, Definitions	Compounds with no kinase above the threshold; ratio off-target cap	`dataset_descriptive_stats.py`
Methods, Ratio; Discussion	Bosc et al. scores and CATDS proxy (Table S2)	`bosc_catds_comparison.py`
Methods, Faithfulness; Figure 1	Published-form entropy and Gini (Table S3); operationalization scatter	`original_operationalization_robustness.py`, `metric_fidelity_robustness.py`
Methods, Rank Stability	Table 1 with mean instead of median ranks	`claim_checks.py` (3)
Methods, Properties; Table 3; D2	Pairs reversing in B, exact for the S-score and on 101 and 201 grid points otherwise	`d3_pair_reversals.py`
Methods, Candidate	Gate and smoothing-width bound for D3, including every β in B	`d4_entropy_vs_candidate.py` (1)
Methods, Candidate	Candidate and entropy under measurement noise	`claim_checks.py` (23)
Table 1	Correlations between definitions (Davis, Klaeger)	`selectivity_analysis.py`, `klaeger_analysis.py`
Results, Two Families	Klaeger correlations without zero-active compounds; Gini bootstrap	`claim_checks.py` (2)
Figure 2; Results, Replication	Correlations on four datasets; replicated instability relations	`additional_datasets_analysis.py` (`cross_dataset_summary.csv`)
Results, Replication	Within-family and ratio ranges, all and active-only compounds	`claim_checks.py` (11)
Results, Replication; D1	Zero-active contrast within each definition's sweep	`claim_checks.py` (13)

Location	Result	Script (output section)
Results, Replication; Type 1; D1	Pooled zero-active contrast, with and without undefined configurations; where each definition ranks zero-active compounds	`claim_checks.py` (20)
Table 2; Figure 3	Profile features vs within-family instability	`instability_by_family.py`
Results, Instability	Overall rank instability, mean and range	`klaeger_analysis.py`, `selectivity_analysis.py`
Results, Instability	Entropy instability vs share of kinases crossing the baseline sweep	`claim_checks.py` (8)
Results, Instability	Range vs instability holding the number of active kinases fixed	`claim_checks.py` (15)
Results, Instability; Section S6	Skewness association and threshold weight	`claim_checks.py` (16)
Results, Instability	Range vs S-score instability on the 46 shared compounds	`claim_checks.py` (17)
Results, Instability	Table 2 with a Bonferroni correction	`claim_checks.py` (21)
Results, Instability	Anastassiadis gap–ratio relation without saturated compounds	`claim_checks.py` (4)
Results, Instability	Undefined skewness and range: placeholder sensitivity	`feature_placeholder_sensitivity.py`
Results, Instability; Section S6	Further candidate features	`instability_predictor_candidates.py`
Results, Sources	The 20 most unstable Klaeger compounds, and the 20 most unstable with an active kinase, by type	`claim_checks.py` (6)
Results, Type 1; Section S6	Instability and median ranks of individual Klaeger compounds	`klaeger_analysis.py` (`klaeger_selectivity_results.csv`)
Results, Type 1	Were the zero-active compounds' targets quantified	`claim_checks.py` (1)
Results, Type 2	Ratio at near-tied top targets	`d2_near_tie_sensitivity.py`
Results, Type 3; Section S6	Ten largest entropy–ratio disagreements	`claim_checks.py` (5)
Section S6	Instability by number of active kinases; most stable compounds	`claim_checks.py` (7, 18)
Figure 4	Binding profiles of three Davis compounds	`binding_profiles.py`
Figure 5; Table S1	Panel-size recovery and p^* at four bars	`panel_size_analysis.py`, `pstar_bar_sensitivity.py`
Results, Panel Size	Gini p^* without compounds whose Gini is undefined	`claim_checks.py` (14)
D1; Figure 6	Zero-active instability under each convention; D1 and D2 panels	`undefined_score_convention.py`, `failure_mode_illustration.py`
Candidate; Table S4	Orientation, D2, D3 at τ^*, smoothing-width sweep	`candidate_validation.py`
Candidate	Agreement with the entropy	`candidate_measure.py`
Candidate	Dependence on the activity threshold τ^*	`claim_checks.py` (22)
Candidate; Section S5	Variants of the candidate and their panel sizes; quantile floor	`family_construction.py`, `family_panelsize.py`

Location	Result	Script (output section)
Recovery; Figure 7	Coverage, Spearman and pair recovery on random subpanels; influence diagnostic	`panel_recovery.py`
Smooth Hinge	Entropy and candidate under one added kinase at τ^*; worked example	`d4_entropy_vs_candidate.py` (2–6)
Smooth Hinge	Realistic panel expansion	`claim_checks.py` (19)
Discussion	Dataset vs definition on the 46 shared compounds, on the shared kinases and on the full panels	`davis_klaeger_overlap.py`, `platform_vs_definition.py`
Discussion	Instability vs cross-dataset rank change	`claim_checks.py` (9)
Discussion	Maximum affinities of the zero-active compounds	`failure_mode_illustration.py`
Discussion	Compounds shared by Klaeger and Anastassiadis	`claim_checks.py` (12)
Discussion, Limitations	Clinical outcomes	`clinical_outcomes_correlation.py`, `faers_pull.py`